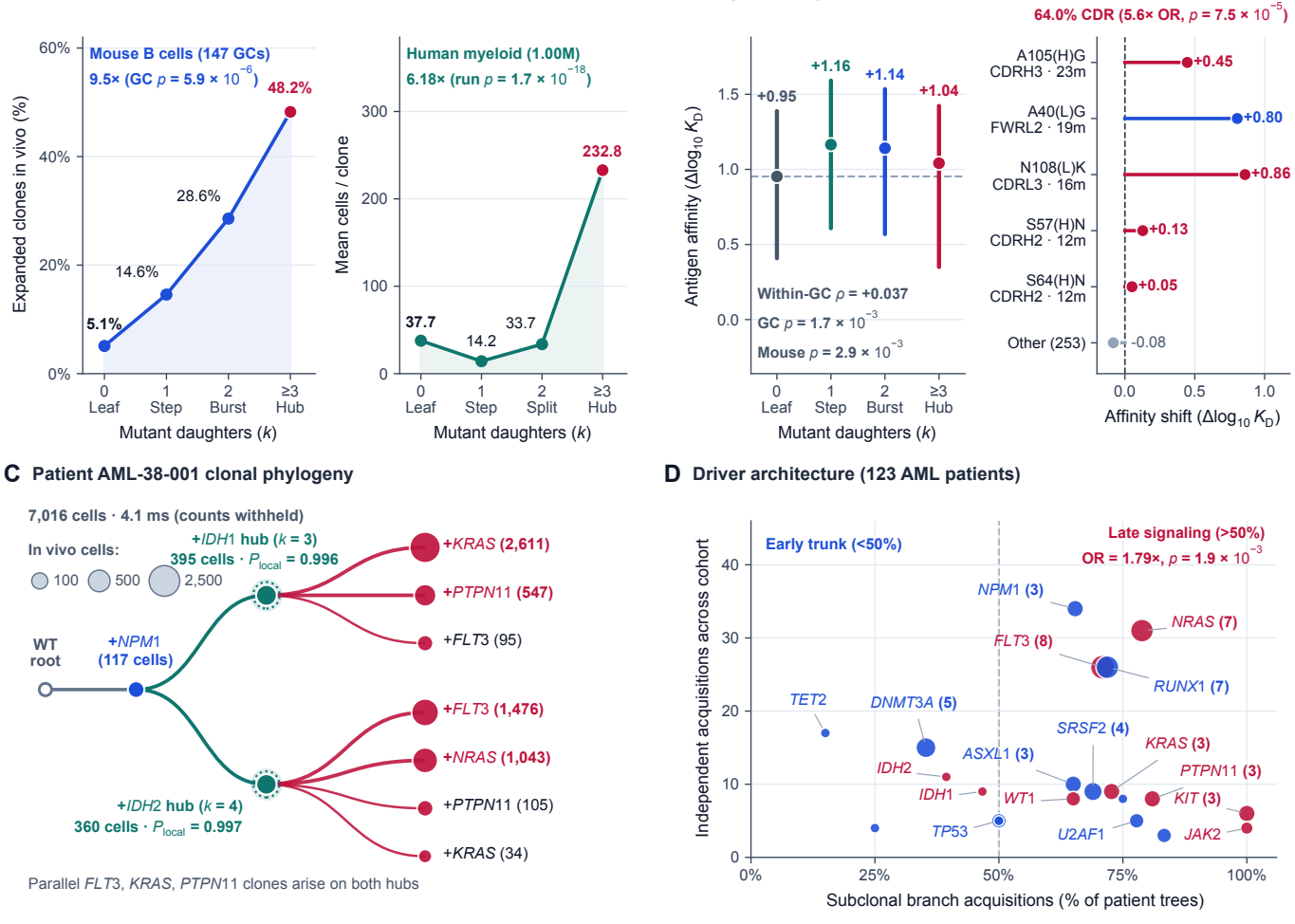

A Clonal expansion without cell counts (1.01M cells)
Mouse B cells (147 GCs)
9.5× (GC p = 5.9 × 10−6)
Expanded clones in vivo (%)
60%
40%
20%
0%
5.1%
14.6%
28.6%
48.2%
0 Leaf
1 Step
2 Burst
≥3 Hub
Mutant daughters (k)
Human myeloid (1.00M)
6.18× (run p = 1.7 × 10−18)
Mean cells / clone
300
200
100
0
37.7
14.2
33.7
232.8
0 Leaf
1 Step
2 Split
≥3 Hub
Mutant daughters (k)
B Antigen affinity & parallel CDR drivers
64.0% CDR (5.6× OR, p = 7.5 × 10−5)
Antigen affinity (Δlog10 KD)
1.5
1.0
0.5
0.0
+0.95
+1.16
+1.14
+1.04
Within-GC p = +0.037
GC p = 1.7 × 10−3
Mouse p = 2.9 × 10−3
0 Leaf
1 Step
2 Burst
≥3 Hub
Mutant daughters (k)
A105(H)G CDRH3 · 23m
+0.45
A40(L)G FWRL2 · 19m
+0.80
N108(L)K CDRL3 · 16m
+0.86
S57(H)N CDRH2 · 12m
+0.13
S64(H)N CDRH2 · 12m
+0.05
Other (253)
-0.08
0.0
0.5
1.0
Affinity shift (Δlog10 KD)
C Patient AML-38-001 clonal phylogeny
7,016 cells · 4.1 ms (counts withheld)
In vivo cells:
100
500
2,500
+IDH1 hub (k = 3)
395 cells · Plocal = 0.996
WT root
+NPM1 (117 cells)
+KRAS (2,611)
+PTPN11 (547)
+FLT3 (95)
+FLT3 (1,476)
+NRAS (1,043)
+PTPN11 (105)
+KRAS (34)
+IDH2 hub (k = 4)
360 cells · Plocal = 0.997
Parallel FLT3, KRAS, PTPN11 clones arise on both hubs
D Driver architecture (123 AML patients)
Early trunk (<50%)
Late signaling (>50%)
OR = 1.79×, p = 1.9 × 10−3
Independent acquisitions across cohort
40
30
20
10
0
NPM1 (3)
NRAS (7)
FLT3 (8)
RUNX1 (7)
TET2
DNMT3A (5)
SRSF2 (4)
KRAS (3)
ASXL1 (3)
IDH2
PTPN11 (3)
IDH1
WT1
KIT (3)
TP53
U2AF1
JAK2
0%
25%
50%
75%
100%
Subclonal branch acquisitions (% of patient trees)